

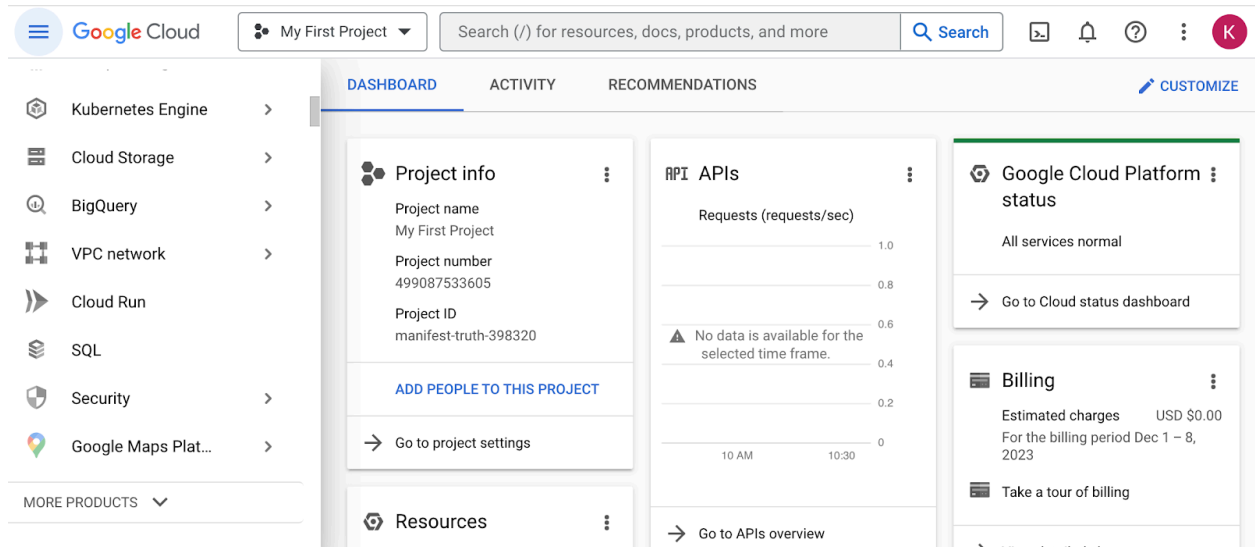


Get started with BigQuery

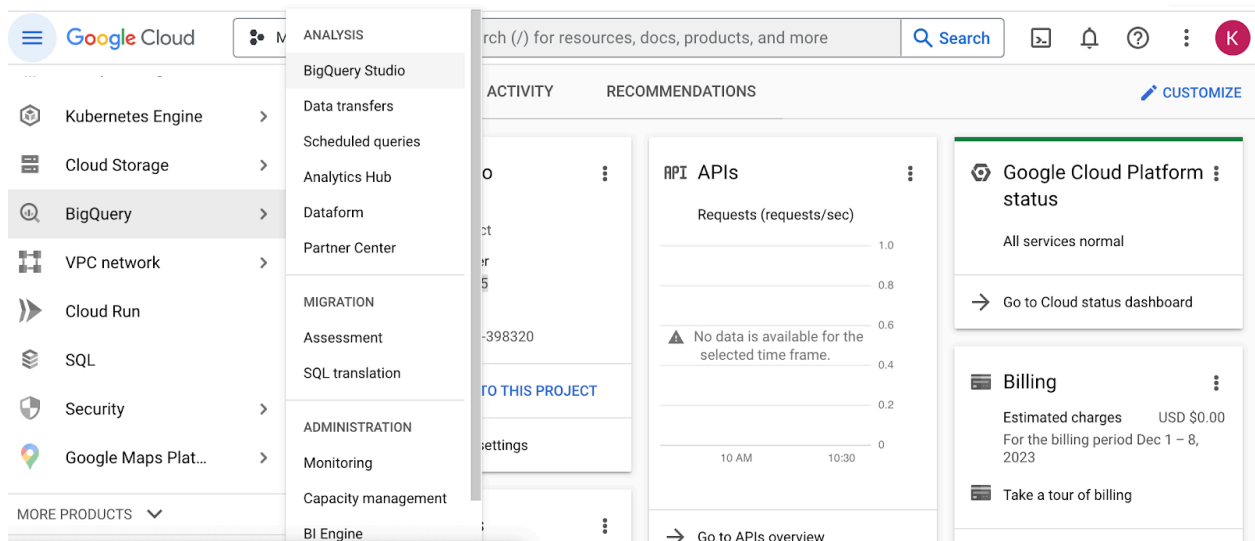
BigQuery is a data warehouse on the Google Cloud Platform used to query and filter large datasets, aggregate results, and perform complex operations. Throughout this program, you're going to use BigQuery to practice your SQL skills and collect, prepare, and analyze data. At this point, you have set up your own account. Now, explore some of the important elements of the SQL workspace. This will prepare you for the upcoming activities in which you will use BigQuery. Note that BigQuery updates its interface frequently, so your console might be slightly different from what is described in this reading. That's okay; use your troubleshooting skills to find what you need!

Log in to BigQuery

When you log in to BigQuery using the landing page, you will automatically open your project space. This is a high-level overview of your project, including the project information and the current resources being used. From here, you can check your recent activity.

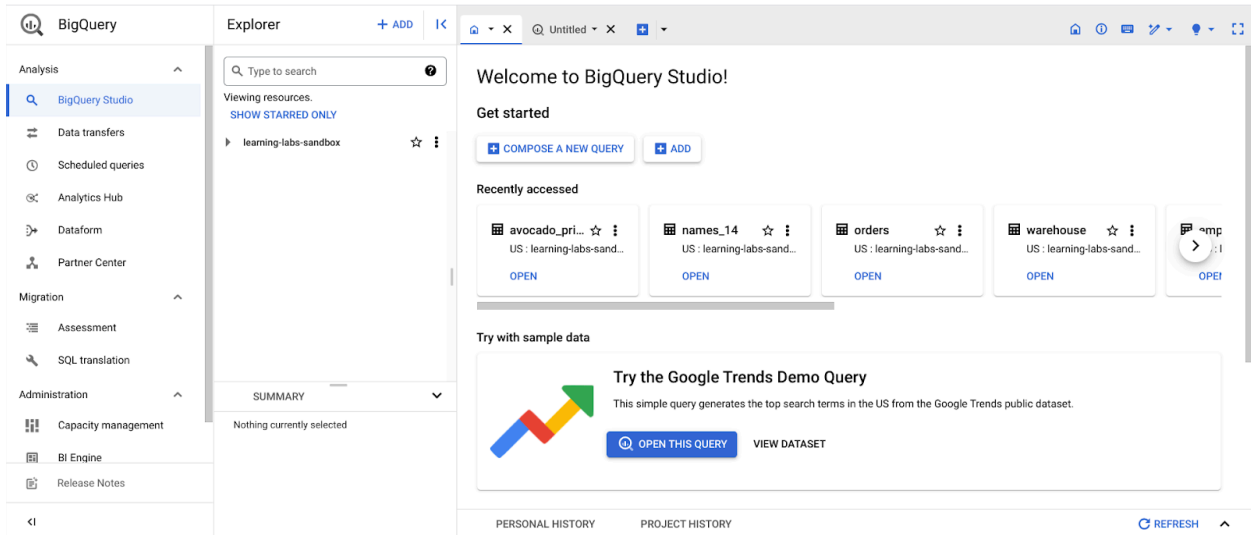


Navigate to your project's BigQuery Studio by selecting BigQuery from the navigation menu and BigQuery Studio from the dropdown menu.



BigQuery Studio components

Once you have navigated to BigQuery Studio by selecting BigQuery from the project space, most of the major components of the BigQuery console will be present: the Navigation pane, the Explorer pane, and the SQL Workspace.



The Navigation pane

On the console page, find the Navigation pane. This is how you navigate from the project space to the BigQuery tool. This menu also contains a list of other Google Cloud Project (GCP) data tools. During this program, you will focus on BigQuery, but it's useful to understand that the GCP has a collection of connected tools data professionals use every day.

The Explorer pane

The Explorer pane lists your current projects and any starred projects you have added to your console. It's also where you'll find the + ADD button, which you can use to add datasets.

Explorer

+ ADD



 Type to search



Viewing resources.

[SHOW STARRED ONLY](#)

▶ manifest-truth-398320



This button opens the Add dialog that allows you to open or import a variety of datasets.

Add



Source

🔍 Search for data sources

Popular sources



Local file

Upload a local file



Google Cloud Storage

Google object storage service



Connections to external data sources

Connection from BigQuery to an external data source

Additional sources

Viewing all 28 results.

Add Public Datasets

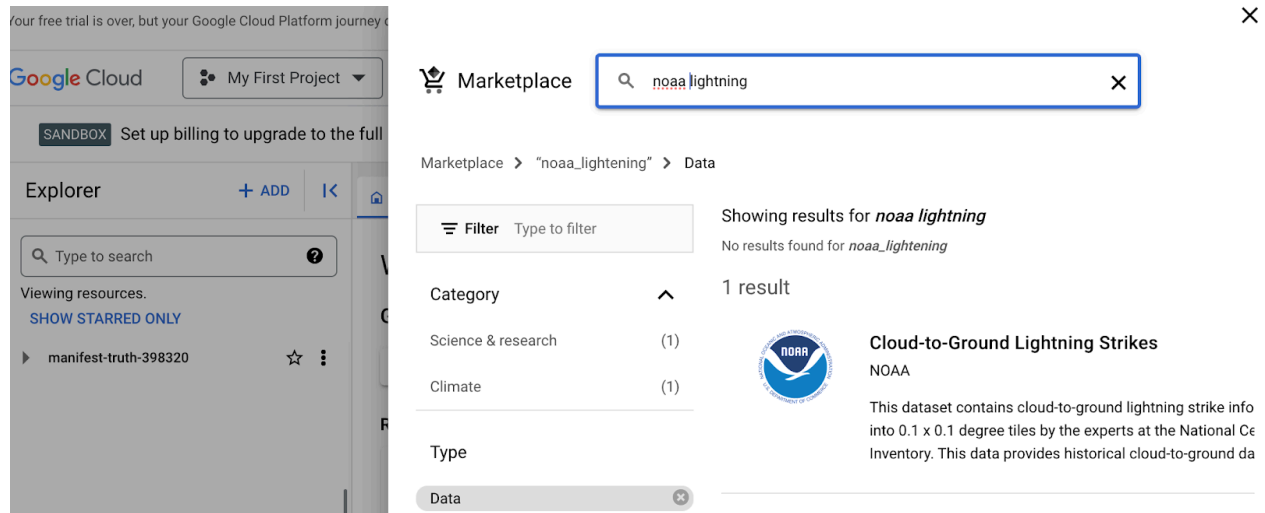
BigQuery offers a variety of public datasets from the Google Cloud Public Dataset Program. Scroll down the Add dialog to the Public Datasets option.



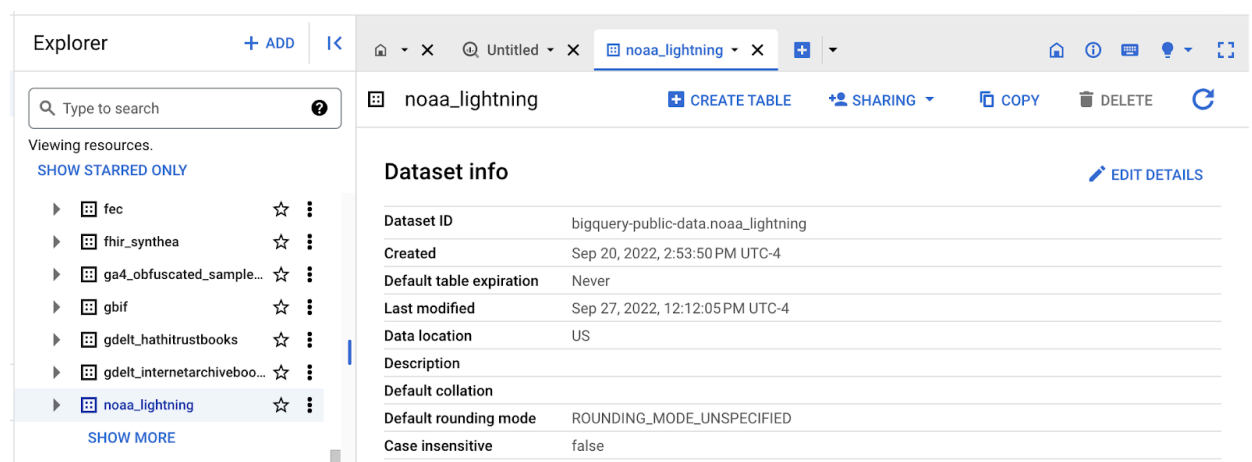
Public Datasets

BigQuery public datasets from the Google Cloud Public Dataset Program

Select Public Datasets. This takes you to the Public Datasets Marketplace, where you can search for and select public datasets to add to your BigQuery console. For example, search for the "noaa lightning" dataset in the Marketplace search bar. When you search for this dataset, you will find NOAA's Cloud-to-Ground Lightning Strikes data.



Select the dataset to read its description. Select View dataset to create a tab of the dataset's information within the SQL workspace.



The Explorer Pane lists the noaa_lightning and other public datasets.

Star and examine Public Datasets

You added the public noaa_lightning dataset to your BigQuery Workspace, so the Explorer pane displays the noaa_lightning dataset, along with the list of other public datasets. These datasets are nested under bigquery-public-data. Star bigquery-public-data by navigating to the top of the Explorer pane and selecting the star next to bigquery-public-data.

Explorer

Type to search

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Viewing resources.

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bigquery-public-data

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 External connections

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 america_health_rankings

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 austin_311

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Starring bigquery-public-data will enable you to search for and add public datasets by scrolling in the Explorer pane or by searching for them in the Explorer search bar.

For example, you might want to select a different public dataset. If you select the second dataset, "austin_311," it will expand to list the table stored in it, "311_service_requests."

Explorer

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austin_311

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311_service_requests

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austin_bikeshare

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The Explorer pane with the “bigquery-public data” and “austin_311” datasets expanded, revealing the “311_service_requests” table

When you select a table, its information is displayed in the SQL Workspace. Select the 311_service_requests table to examine several tabs that describe it, including:

- Schema, which displays the column names in the dataset
- Details, which contains additional metadata, such as the creation date of the dataset
- Preview, which shows the first rows from the dataset

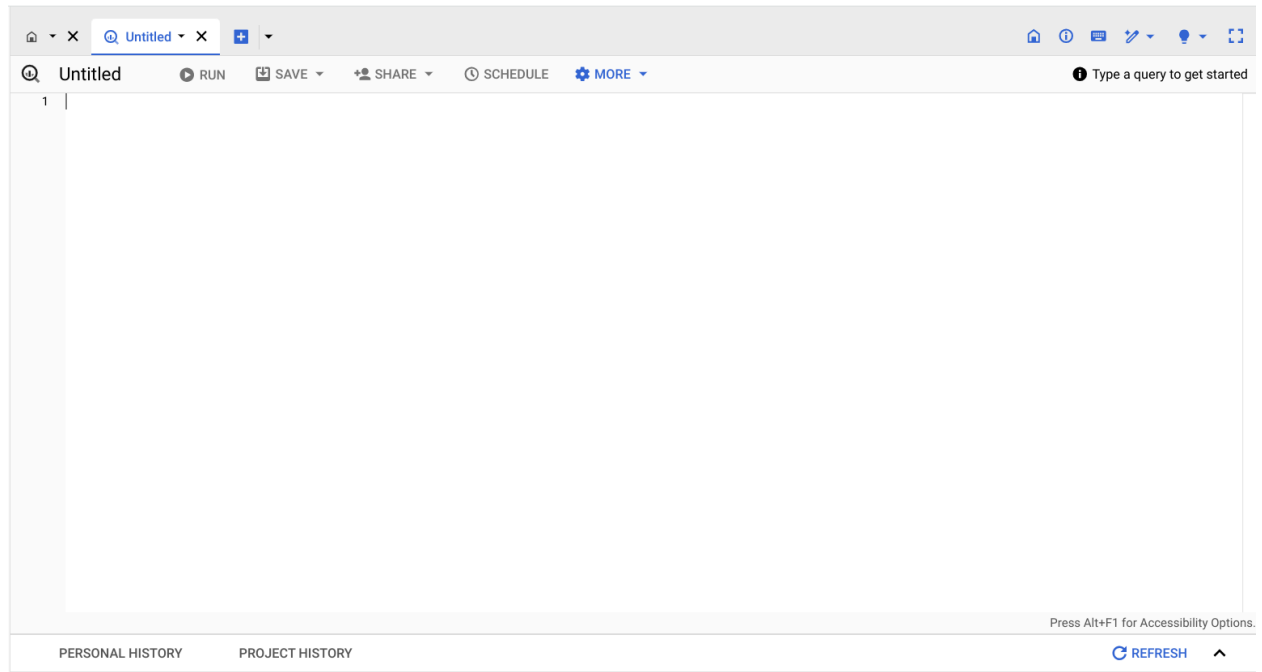
The screenshot shows the Google Cloud BigQuery Explorer interface. On the left is the 'Explorer' pane with a search bar and a tree view of resources. The '311_service_requests' dataset is selected. The main pane on the right shows the '311_service_r...' dataset with tabs for 'SCHEMA', 'DETAILS', 'PREVIEW', 'LINEAGE', and 'DATA PRO'. The 'SCHEMA' tab is active, displaying a table with columns: 'Field name', 'Type', and 'Mode'. The table lists three fields: 'UNIQUE_KEY' (STRING, NULLABLE), 'COMPLAINT_DESCRIPTION' (STRING, NULLABLE), and 'SOURCE' (STRING, NULLABLE).

Field name	Type	Mode
UNIQUE_KEY	STRING	NULLABLE
COMPLAINT_DESCRIPTION	STRING	NULLABLE
SOURCE	STRING	NULLABLE

Additionally, you can select the Query button from the menu bar in the SQL Workspace to query this table.

The SQL Workspace

The final menu pane in your console is the SQL Workspace. This is where you will actually write and execute queries in BigQuery.



The SQL Workspace also gives you access to your personal and project history, which stores a record of the queries you've run. This can be useful if you want to return to a query to run it again or use part of it in another query.

Upload your data

In addition to offering access to public datasets, BigQuery also gives you the ability to upload your own data directly into your workspace. Access this feature by opening the + ADD menu again or by clicking the three vertical dots next to your project's name in the Explorer pane. This will give you the option to create your own dataset and upload your own tables. You will have the opportunity to upload your own data in an upcoming activity to practice using this feature!

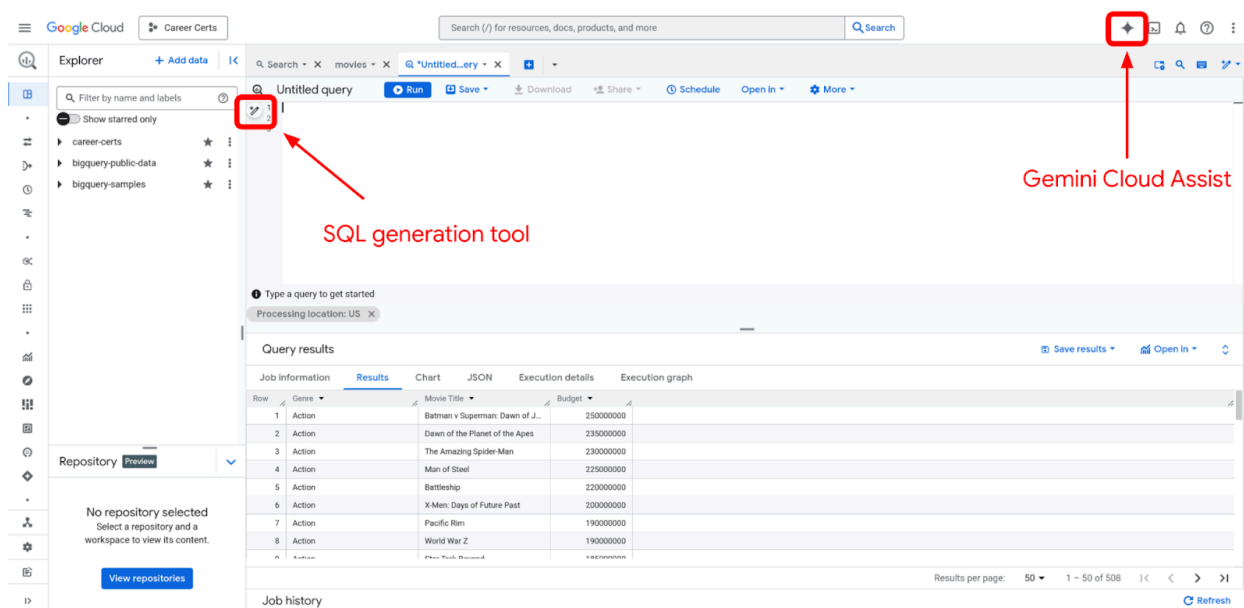


Within BigQuery you'll also find powerful AI tools to help you with your code—tools like Gemini Cloud Assist and the SQL generation tool. These tools function as assistants that can help analyze code and find errors, suggest modifications, and also interact in a conversational way to answer many other questions, whether they're technical or more conceptual.

To enable Gemini Cloud Assist, find the Gemini icon in the upper-right corner of the BigQuery console to toggle a Gemini Cloud Assist panel. Here, you can query Gemini using natural language.

To use the SQL generation tool within the SQL editor itself, locate the magic wand icon at the upper-left corner of the SQL query window. Clicking here will open a window where you can ask Gemini to generate SQL queries for your data.

While features like AI-powered SQL generation, code completion, and data insights within BigQuery can accelerate your workflow, please remember that this technology is still evolving. The suggestions, code, or explanations provided may not always be perfectly accurate, optimal, or secure. Always review and validate any AI-generated output before executing queries or relying on the information. Treat the AI assistance as a helpful co-pilot, but maintain oversight and responsibility for your final queries and analysis.



For more information on these tools and detailed walkthroughs refer to the following pages:

[Gemini in BigQuery overview](#)

[Write queries with Gemini assistance](#)

Key takeaways

BigQuery's SQL workspace allows you to search for public datasets, run SQL queries, and even upload your own data for analysis. Whether you're working with public datasets, running SQL queries, or uploading your own data, BigQuery's SQL workspace offers a range of features to support all kinds of data analysis tasks. Throughout this program, you will be using BigQuery to practice your SQL skills, so being familiar with the major components of your BigQuery console will help you navigate it effectively in the future!